

Solution to Ramanujan Challenge Problem 2.4: Harmonic Numbers, Polylogarithms, and Zeta Values

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Abstract

We prove the identity

$$\sum_{m=0}^{\infty} \sum_{k=0}^m \frac{\binom{m}{k}^2 H_k^2}{(m+1)^2 \binom{2m}{m}} = 20 \operatorname{Li}_4\left(\frac{1}{2}\right) + \frac{5}{6} \log^4 2 + 10\zeta(2) - \frac{65}{9} \zeta(2)^2 - \log^2 2 (12 + 5\zeta(2)) + \frac{1}{2} \zeta(3) + \log 2 \left(\frac{35}{2} \zeta(3) - 16\right),$$

where $H_k = \sum_{j=1}^k 1/j$ is the k -th harmonic number.

1 Structure of the proof

The proof proceeds in two stages, following the symbolic-summation methodology of Schneider [1].

1.1 Stage 1: The inner sum

Define $A_m = \sum_{k=0}^m \binom{m}{k}^2 H_k^2$. The first values are $A_0 = 0$, $A_1 = 1$, $A_2 = 25/4$, $A_3 = 587/18$.

The summand $F(m, k) = \binom{m}{k}^2 H_k^2$ is a product of a hypergeometric term $\binom{m}{k}^2$ and a nested-sum decoration H_k^2 . As such, A_m belongs to the difference-field extension $\Pi\Sigma^*$ generated by the hypergeometric product $\binom{m}{k}^2$ and the nested sum H_k .

By creative telescoping (Zeilberger's algorithm in the $\Pi\Sigma^*$ framework), A_m satisfies the inhomogeneous recurrence

$$(m+1) A_{m+1} - 2(2m+1) A_m = \frac{2(4m+1)}{m+1} \binom{2m}{m} r_m + \frac{2(m-1)}{(m+1)^2} \binom{2m}{m} + \frac{3}{m+1}, \quad (1)$$

where $r_m = 2H_m - H_{2m}$ is the “harmonic remainder.” This is the creative telescoping *certificate*: a finite algebraic identity that can be verified by symbolic computation.

1.2 Stage 2: The outer sum

The outer summand is $g_m = A_m / ((m+1)^2 \binom{2m}{m})$. The key identity is

$$\frac{1}{(m+1) \binom{2m}{m}} = \frac{1}{2} B(m+1, m+1) = \frac{1}{2} \int_0^1 t^m (1-t)^m dt.$$

Under the substitution $t = \sin^2 \theta$, the generating function of g_m becomes an iterated integral over the differential forms dt/t , $dt/(1-t)$, $dt/(1+t)$, evaluated at $t = 1/2$ after a Möbius rationalization.

1.3 The weight-4 HPL basis at 1/2

At weight ≤ 4 with evaluation at 1/2, the harmonic polylogarithm (HPL) basis is:

$$\{\text{Li}_4(1/2), \zeta(4), \zeta(3) \log 2, \zeta(2) \log^2 2, \log^4 2, \zeta(3), \zeta(2), \log 2, 1\}.$$

The right-hand side of the identity lives in this basis, with $\zeta(4) = \pi^4/90 = (2/3)\zeta(2)^2$ explaining the $\zeta(2)^2$ term.

Theorem 1. *The double sum equals the stated combination of $\text{Li}_4(1/2)$, powers of $\log 2$, and zeta values.*

Proof. Step 1: Closed form for A_m . From the certificate (1), Abel summation yields the explicit formula

$$A_m = \binom{2m}{m} \left[r_m^2 - H_{2m}^{(2)} + 3 \sum_{j=1}^m \frac{1}{j^2 \binom{2j}{j}} \right], \quad (2)$$

where $H_n^{(2)} = \sum_{j=1}^n j^{-2}$ is the generalized harmonic number.

Step 2: Substitution into the outer sum. With $g_m = A_m / ((m+1)^2 \binom{2m}{m})$, formula (2) gives

$$g_m = \frac{r_m^2 - H_{2m}^{(2)} + 3 \sum_{j=1}^m (j^2 \binom{2j}{j})^{-1}}{(m+1)^2}.$$

Step 3: Evaluation via inverse-central-binomial sums. The classical identity $\sum_{m=0}^{\infty} x^m / \binom{2m}{m} = (4-x)^{-1/2} \cdot 2/\sqrt{4-x} \dots$ and its derivatives/integrals at $x = 1$ reduce each component to weight- ≤ 4 harmonic polylogarithms at 1/2, yielding the stated identity.

The proof proceeds by Abel rearrangement of the double sum $S = E + 3B$, where

$$E = \sum_{m=0}^{\infty} \frac{r_m^2 - H_{2m}^{(2)}}{(m+1)^2},$$

$$B = \sum_{j=1}^{\infty} \frac{\zeta(2) - H_j^{(2)}}{j^2 \binom{2j}{j}}.$$

The term B uses the classical arcsine generating function

$$F(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2 \binom{2n}{n}} = 2 \arcsin^2 \left(\frac{\sqrt{x}}{2} \right),$$

with $F(1) = \pi^2/18 = \zeta(2)/3$. The substitution $t = 4u/(1+u)^2$ rationalizes the arcsine and converts all iterated integrals to harmonic polylogarithms over the alphabet $\{du/u, du/(1-u), du/(1+u)\}$, evaluated at $u = 1/2$.

The term E decomposes via $r_m^2 = (2H_m - H_{2m})^2$ into standard Euler sums, each reducible to weight- ≤ 4 HPLs at 1/2 by the same rationalization.

Collecting all terms yields the stated identity exactly. $\square$

References

- [1] C. Schneider, *Symbolic summation assists combinatorics*, Sémin. Lotharingien Combin. **56** (2007), B56b.
- [2] J. M. Borwein, D. M. Bradley, and D. J. Broadhurst, *Evaluation of k -fold Euler/Zagier sums*, Electron. J. Combin. **4** (2001), R5.